MD TANVIR ALAM

Chief Technology Officer | Alawaf Technologies Limited

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GitHub: github.com/tanvir-ux | Portfolio: tanvir-ux.github.io | LinkedIn: linkedin.com/in/mdtanviralam

Open to: Remote from Bangladesh, or APAC (Singapore EP, Jakarta, India)

SUMMARY

Chief Technology Officer at Alawaf Technologies Limited and a hands-on software engineer with 5+ years designing distributed systems, high-concurrency banking backends, and production Agentic AI platforms. Builds generative AI applications, LLM solutions, agentic workflows, RAG pipelines, and knowledge graphs, and designs Java / Spring Boot microservices, REST APIs, and Kafka pipelines. Expert in Java, Spring Boot, Microservices, REST, Kafka, Kubernetes, Docker, AWS, PostgreSQL, Oracle, CI/CD, System Design, Distributed Systems, LangChain, RAG, and MCP (Model Context Protocol). Recently landed substantive fixes in Meta Infer, Google Gson, and Google Auto. Previously worked as a contractor customizing Oracle FLEXCUBE and Oracle Banking Payments (OBPM) for Bank Danamon (Indonesia) and SMBC (Singapore); not an Oracle employee.

SKILLS

Languages and Frameworks: Java, Spring Boot, Microservices, REST API Design, JVM, Grails, Node.js, Python, FastAPI, OpenAPI

Architecture and Patterns: Event-Driven Architecture, Design Patterns, DDD, CQRS, Event Sourcing, Clean Architecture, SOLID, TDD, Agile/Scrum, Distributed Systems, System Design, CI/CD

AI and Machine Learning: Agentic AI, Model Context Protocol (MCP), FastMCP, LangChain, RAG, OpenAI API, Anthropic Claude, Ollama, Qwen, Gemini, Prompt Engineering, Keras, TensorFlow, PyTorch

Databases and Vector Stores: Oracle PL/SQL (CDB/PDB), PostgreSQL, MySQL, Redis, Flyway, pgvector, ChromaDB, FAISS, PySpark, Databricks

Cloud and DevOps: AWS (EKS, ECS, Lambda, S3), GCP, Azure, Docker, Kubernetes, Apache Kafka, MQTT, VerneMQ, Nginx, GitHub Actions

Banking and Security: Oracle FLEXCUBE, Oracle Banking Payments (OBPM), SWIFT, OAuth2, JWT

EXPERIENCE

Chief Technology Officer | Alawaf Technologies Limited — *Present*

- Build generative AI applications, LLM solutions, agentic workflows, RAG pipelines, and knowledge graphs.
- Work on distributed systems and Java / Spring backends used in regulated environments.

Chief Technology Officer and Senior Software Engineer | Jonlab LLC — *Nov 2025 – Present*

- Define long-term technical vision and strategic roadmaps, orchestrating cross-functional engineering teams to scale backend systems.
- Direct the design and production integration of enterprise Agentic AI pipelines and custom LLM interfaces using LangChain, RAG, and MCP.
- Spearhead modernization of legacy business layers into high-availability Microservices on Java, Spring Boot, Kubernetes, Kafka, and cloud infrastructure.
- Optimize system stability and server performance across distributed services.

Lead Software Engineer / AI Consultant | MehvozSoft — *Aug 2025 – Oct 2025*

- Designed and developed self-executing AI agents with LangChain, orchestrating multi-step reasoning to reduce manual oversight by 40%.
- Engineered RAG architectures so private language models can query internal document repositories and call external APIs via tool-calling.
- Integrated resilient messaging pipelines with Kafka and Redis to streamline background processing and improve Java / Spring Boot performance.

Senior Software Engineer (Java) | Midas Software Solutions Ltd. — *Mar 2023 – Aug 2025*

- **Bank Danamon (Indonesia):** Architected and executed migration of a core digital banking monolith to a distributed Spring Boot Microservices structure, delivering 70% faster deployment cycles for high-concurrency instances.
- Designed, optimized, and maintained over 100 microservice REST endpoints using OAuth2 and JWT, supporting 50k+ concurrent users.
- As a contractor (not an Oracle employee), customized Oracle FLEXCUBE and Oracle Banking Payments (OBPM) for Bank Danamon (Indonesia) and SMBC (Singapore) using JavaScript, XML, and Oracle PL/SQL to execute cross-border SWIFT transactions.
- Applied System Design and Distributed Systems practices (Kafka, Kubernetes, Docker, CI/CD) on production banking workloads with PostgreSQL and Oracle.

Application Developer | MehvozSoft — *Sep 2021 – Feb 2023*

- Developed and deployed cross-platform mobile utilities (including Express Entry Calculator and Adhaan applications) on iOS and Android using Ionic and Angular.
- Crafted database schemas and optimized SQL queries, indexing, and procedures on PostgreSQL and MySQL to accelerate transactional execution.

SELECTED PROJECTS

Original portfolio demos, not employer or client production systems. Repositories: github.com/tanvir-ux

ai-engineering-stack *(replay) | Knowledge-copilot demo across UI, sign-in, API, agent workflow, retrieval, and deployment.*

agent-platform | *One support task across several agent frameworks, with FastAPI, MCP, and A2A. Demo.*

graphrag-service | *GraphRAG and vector RAG, with an evaluation harness. Demo.*

llm-finetune-serve | *CPU LoRA fine-tune of SmolLM2 and an OpenAI-compatible server. Demo.*

agentic-mcp-gateway | *Python, FastAPI, MCP, RAG, Docker.*
MCP server plus FastAPI HTTP agent sharing one tool registry for document retrieval, payment searches, and ticket creation. ReAct loop with CI stubbing and configurable live LLM providers. The default path runs without an API key.

banking-payments-spring | *Java 17, Spring Boot 3.3, Kafka, PostgreSQL, Docker.*
Credit-transfer REST microservice with JWT resource security, idempotent POST, JPA persistence, an AML sanctions stub, and ISO 20022 / SWIFT-inspired Kafka domain events.

kafka-spring-microservices | *Java, Spring Boot, Kafka, Kubernetes, PostgreSQL.*
Transfer, ledger, and notifications services with double-entry execution, idempotent consumers, publish-after-commit events, plus Docker Compose and Kubernetes manifests.

OPEN SOURCE

Verified substantive fixes. Links go to the original repositories.

- **Meta Infer:** Track streams from `Socket.getInputStream` and `getOutputStream` in Pulse (PR #2302).
- **Meta Infer:** Track `Deflater` and `Inflater` as native resources in Pulse (PR #2275).
- **Google Gson:** Record the JSON path after `promoteNameToValue` (PR #3098).
- **Google Auto:** Copy `javax` and `jakarta` `@Qualifier` onto generated AutoFactory parameters (commit d10c050).

EDUCATION

Stamford University Bangladesh — B.Sc. in Electrical and Electronic Engineering (EEE) *Graduated Aug 2021 | CGPA: 3.25*

Machine Learning Specialization — Stanford University and DeepLearning.AI (Coursera) *Credential ID: 4UGJY8KESNCZ*